



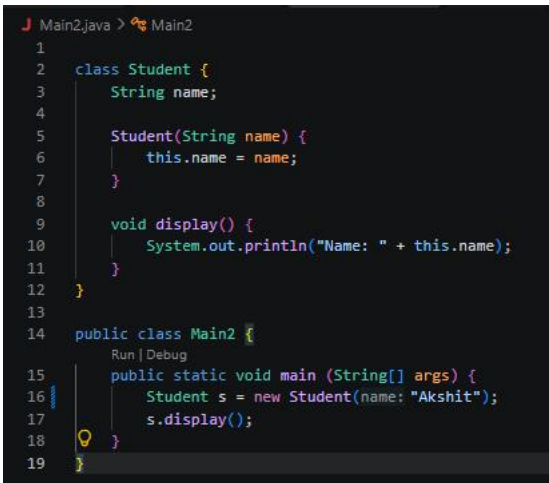
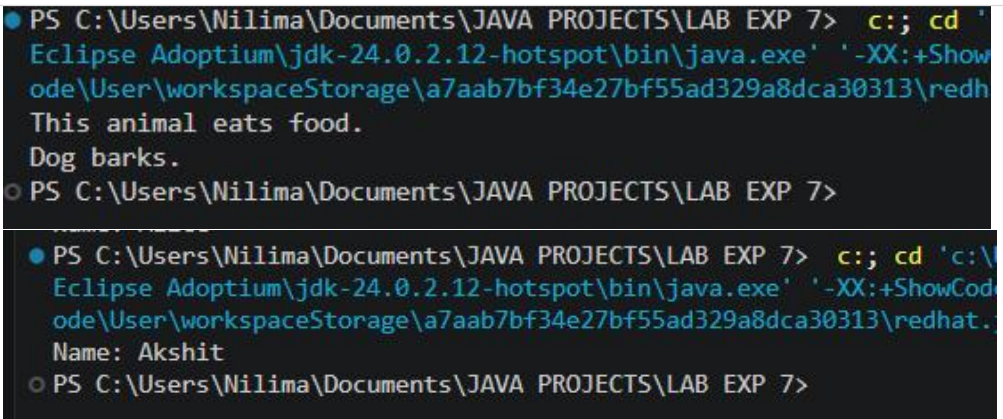
## Symbiosis Institute of Technology, Pune

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

### Lab Assignment No.: 7

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name of Student</b>              | Akshit Gawade                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>PRN No.</b>                      | 25070122029                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Batch</b>                        | 2025-2029                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Class</b>                        | A1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Academic Year &amp; Semester</b> | Second Year<br>Semester III                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Date of Submission</b>           | 11/7/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Title of Assignment:</b>         | Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Theory</b>                       | Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming |
| <b>Program Code</b>                 | <p>1.</p> <pre>1 2 class Animal { 3     void eat () { 4         System.out.println("This animal eats food."); 5     } 6 } 7 8 class Dog extends Animal { 9     void bark() { 10        System.out.println("Dog barks."); 11    } 12 } 13 14 public class Main1 { 15     public static void main (String[] args) { 16         Dog d = new Dog(); 17         d.eat(); 18         d.bark(); 19     } 20 }</pre>                                                                         |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>2.</p>  <pre> J Main2.java &gt; Main2 1 2 class Student { 3     String name; 4 5     Student(String name) { 6         this.name = name; 7     } 8 9     void display() { 10        System.out.println("Name: " + this.name); 11    } 12 } 13 14 public class Main2 { 15     public static void main (String[] args) { 16         Student s = new Student(name: "Akshit"); 17         s.display(); 18     } 19 } </pre>                                                                                                                                                     |
| Output Screenshots |  <pre> PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 7&gt; c::; cd 'C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 7' PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 7&gt; java -XX:+ShowCode -XX:+ShowCode -XX:+ShowCode -XX:+ShowCode -XX:+ShowCode This animal eats food. Dog barks. PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 7&gt; PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 7&gt; java -XX:+ShowCode -XX:+ShowCode -XX:+ShowCode -XX:+ShowCode -XX:+ShowCode Name: Akshit PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 7&gt; </pre> |
| Exercise Qns       | <p>1. Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

```

Exp1.java > Exp1 > main(String[])
1
2 class Shape {
3     void display() {
4         System.out.println(x: "Created a new shape:");
5     }
6 }
7
8 class Circle extends Shape {
9     double area (double r) {
10         return 3.14 * r * r;
11     }
12 }
13
14 class Rectangle extends Shape {
15     double area (double l, double b) {
16         return l * b;
17     }
18 }
19
20 public class Exp1 {
21     public static void main (String[] args) {
22         Circle c = new Circle();
23         Rectangle r = new Rectangle();
24
25         c.display();
26         System.out.println("Area of Circle : " + c.area(r: 10));
27         r.display();
28         System.out.println("Area of Rectangle : " + r.area(l: 20, b: 20));
29     }
30 }

```

2. Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

```

1 interface Product {
2     void displayCategory();
3 }
4
5 class ProductDetails {
6     int id;
7     String name;
8     double price;
9
10    ProductDetails(int id, String name, double price) {
11        this.id = id;
12        this.name = name;
13        this.price = price;
14    }
15
16    void displayDetails() {
17        System.out.println("Product ID : " + id);
18        System.out.println("Product Name : " + name);
19        System.out.println("Price : " + price);
20    }
21 }
22
23 class Electronic extends ProductDetails implements Product {
24     Electronic(int id, String name, double price) {
25         super(id, name, price);
26     }
27
28     public void displayCategory() {
29         System.out.println(x: "Category : Electronic");
30     }
31 }
32
33 class Clothing extends ProductDetails implements Product {
34     Clothing(int id, String name, double price) {
35         super(id, name, price);
36     }
37
38     public void displayCategory() {
39         System.out.println(x: "Category : Clothing");
40     }
41 }

```

```

43 class Grocery extends ProductDetails implements Product {
44     Grocery(int id, String name, double price) {
45         super(id, name, price);
46     }
47
48     public void displayCategory() {
49         System.out.println(x: "Category      : Grocery");
50     }
51 }
52
53 public class Exp2 {
54     Run | Debug
55     public static void main(String[] args) {
56         Electronic e = new Electronic(id: 201, name: "Mobile", price: 45000);
57         Clothing c = new Clothing(id: 202, name: "Shirt", price: 999);
58         Grocery g = new Grocery(id: 203, name: "Milk", price: 65);
59
60         e.displayDetails();
61         e.displayCategory();
62
63         System.out.println();
64
65         c.displayDetails();
66         c.displayCategory();
67
68         System.out.println();
69
70         g.displayDetails();
71         g.displayCategory();
72     }
73 }

```

## Output Screenshots

```

● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 7> c:; cd 'c:\U
Eclipse Adoptium\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCode
ode\User\workspaceStorage\a7aab7bf34e27bf55ad329a8dca30313\redhat.j
Created a new shape:
Area of Circle : 314.0
Created a new shape:
Area of Rectangle : 400.0
○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 7>

```

```

● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 7> c:; cd 'c:\U
Eclipse Adoptium\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCode
ode\User\workspaceStorage\a7aab7bf34e27bf55ad329a8dca30313\redhat.java\
Product ID      : 201
Product Name    : Mobile
Price           : 45000.0
Category        : Electronic

Product ID      : 202
Product Name    : Shirt
Price           : 999.0
Category        : Clothing

Product ID      : 203
Product Name    : Milk
Price           : 65.0
Category        : Grocery
○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 7>

```

## Conclusion

In this experiment, we implemented the concepts of inheritance and interfaces. We learnt the use of keyword **extends** and **implements**.